SETS RECONSTRUCTIBLE WITH MEDIAL AXIS: A GENERALISATION TO RIEMANNIAN MANIFOLDS

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Abstract. We extend the main reconstruction theorem of Białożyt [1] — which characterises exactly which points of X^c are recoverable from the medial axis of a closed set $X \subset \mathbb{R}^n$ — to complete Riemannian manifolds. The key replacements are: half-spaces by *horoballs* (sublevel sets of Busemann functions of rays), and the convex hull by the *horoball hull* $\widehat{X}$ (complement of the union of all X-free open horoballs). The *sufficiency* direction splits into two results. *Theorem A* (no curvature assumption, §3) states that every point of $\widehat{X} \setminus X$ is reconstructible; on compact manifolds, which have no rays, this immediately implies that X^c is always reconstructible. *Theorem B* (Hadamard manifolds, §4) states that every point in a defective maximal X-free horoball is also reconstructible; defectiveness is defined exactly as in the Euclidean case but with horospheres replacing supporting hyperplanes. The *necessity* direction is left as a conjecture (§5), with a partial result and an explicit account of why the same gap already exists in the original Euclidean proof. We close (§7) with an audit of three defects in [1]: a step in Proposition 3 that skips a key inequality, a genuine quantitative gap in Proposition 4 covering small heights, and an inconsistency between $\operatorname{conv} X$ and $\overline{\operatorname{conv}} X$.

Contents

Conventions. Unless stated otherwise, (M^n, g) denotes a complete, connected Riemannian manifold, $X \subsetneq M$ is a non-empty closed subset, $d := d(\cdot, X)$ the distance function to X, and $m(y)$ the (non-empty, by Hopf–Rinow) set of feet of y in X.

1. Motivation and the Euclidean benchmark

Białożyt's Theorem 6 characterises reconstructible points of $X^c := \mathbb{R}^n \setminus X$ for a closed set X as precisely those lying in

$$(1) \qquad \overline{\operatorname{conv} X} \;\cup\; \bigcup \big\{ L^+ : L \text{ supporting hyperplane of } \operatorname{conv} X,\ X \cap L \neq \operatorname{conv} X \cap L \big\},$$

where L^+ is the open half-space on the side of L disjoint from $\operatorname{conv} X$. A *defective* supporting hyperplane is one where $X \cap L \subsetneq \operatorname{conv} X \cap L$; the trapped witness $w \in (\operatorname{conv} X \cap L) \setminus X$ is the analogue of non-convexity of X at that face.

Three features of the Euclidean setting that will require replacement:

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(i) Half-spaces are exactly the *horoballs* of $\mathbb{R}^n$: the sub-level sets $\{b_\gamma < c\}$ where $b_\gamma(y) = -\langle y - \gamma(0), \dot\gamma\rangle$ is the Busemann function of the ray γ.
(ii) The *convex hull* is the intersection of all half-spaces containing X, equivalently the complement of all X-free open half-spaces.
(iii) *Defectiveness* of a half-space L^+ is the condition $(\partial L \cap \overline{\operatorname{conv} X}) \setminus X \neq \emptyset$.

On a curved manifold only item (i) generalises cleanly to all complete manifolds; items (ii) and (iii) require curvature assumptions. The geodesic convex hull is the wrong replacement: on S^n, two generic points already hull to the whole sphere, while the horoball hull stays meaningful.

2. Definitions

Definition 2.1 (Medial axis, reconstructible set)**.** The *medial axis* of X is

$$M_X := \big\{a \in X^c : a \text{ is the base of at least two distinct minimising geodesics to } X\big\}.$$

(Geodesics, not feet, to handle poles: the antipode on S^n of a singleton has one foot and infinitely many geodesics.) The *reconstructible set* is $\mathcal{R}_X := \bigcup_{a\in M_X} B(a, d(a))$.

Remark 2.2. Via local semiconcavity of d on X^c, the set M_X is exactly the non-differentiability locus of d. On Hadamard manifolds (no conjugate points) this coincides with the multi-foot definition.

Definition 2.3 (Busemann function, horoball)**.** For a ray $\gamma : [0,\infty) \to M$ (unit-speed, globally minimising), the *Busemann function* is

$$b_\gamma(y) := \lim_{t\to\infty} \big(d(y, \gamma(t)) - t\big).$$

The limit exists since $t \mapsto d(y, \gamma(t)) - t$ is non-increasing (triangle inequality) and bounded below. b_γ is 1-Lipschitz and $b_\gamma(\gamma(s)) = -s$. Set

$$c_\gamma := \inf_X b_\gamma \ \in\ [-\infty, \infty).$$

When c_γ is finite the *maximal X-free horoball* is

$$\mathcal{H}_\gamma := \{b_\gamma < c_\gamma\}, \qquad S_\gamma := \{b_\gamma = c_\gamma\} \text{ (the bounding horosphere).}$$

Remark 2.4. c_γ is always finite (since $b_\gamma(y) \to -\infty$ along γ but $X \neq \emptyset$), and $\mathcal{H}_\gamma$ is always X-free. If $c_\gamma = -\infty$ (ray passes through X) there is no nontrivial free horoball in that direction.

Definition 2.5 (Horoball hull, defective horoball)**.** The *horoball hull* of X is

$$\widehat{X} := M \setminus \bigcup\big\{X\text{-free open horoballs}\big\} = \big\{y \in M : b_\gamma(y) \geq c_\gamma \text{ for every ray } \gamma\big\}.$$

This is a closed set containing X. A maximal free horoball $\mathcal{H}_\gamma$ is *defective* if $\big(S_\gamma \cap \widehat{X}\big) \setminus X \neq \emptyset$. Such a point $w \in (S_\gamma \cap \widehat{X}) \setminus X$ is a *trapped witness.*

Remark 2.6 (Sanity checks)**.** In $\mathbb{R}^n$: $b_\gamma(y) = -\langle y - \gamma(0), \dot\gamma\rangle$, horoballs are open half-spaces, $\widehat{X} = \overline{\operatorname{conv}} X$, defectiveness recovers the paper's condition (with closure, see §7). On compact M: no rays exist, so $\widehat{X} = M$ and every closed set is trivially defect-free. On $\mathbb{H}^2$ with $X = \{x_1, x_2\}$: $\widehat{X}$ is the lens bounded by the two horocycle arcs through both points, strictly larger than the geodesic segment.

The following is the one non-trivial external input.

Lemma 2.7 (Cut locus density, cf. [2, 3])**.** *Let M be complete and X closed. The* cut locus $\operatorname{Cut}(X)$ *— the set of endpoints of maximal minimising segments of normal geodesics from X — satisfies* $\operatorname{Cut}(X) \subset \overline{M_X}$.

3. Reconstructing the horoball hull

Theorem 3.1 (Hull reconstruction, curvature-free)**.** *Let M be complete and X closed. Every $p \in \widehat{X} \setminus X$ is reconstructible. Equivalently, every non-reconstructible point of X^c lies in an X-free open horoball.*

Proof. Set $\delta := d(p) > 0$. If $p \in M_X$ then $p \in B(p, \delta) \subset \mathcal{R}_X$ and we are done. Otherwise there is a unique minimising geodesic from the foot $x_p = \gamma(0)$ to $p = \gamma(\delta)$; extend to a ray (if the cut time is > 0, which we will establish) by the completeness of M.

Define

$$r := \sup\{t > 0 : B(\gamma(t), t) \cap X = \emptyset\}.$$

The set is an interval containing $(0, \delta]$. If $B(\gamma(t_0), t_0)$ is free and $0 < t < t_0$, then for any $y \in B(\gamma(t), t)$:

$$d(y, \gamma(t_0)) \leq d(y, \gamma(t)) + (t_0 - t) < t + (t_0 - t) = t_0,$$

so $B(\gamma(t), t)$ is free too. For $0 < t \leq \delta$, if $y \in X \cap B(\gamma(t), t)$ then

$$d(p, y) \leq d(p, \gamma(t)) + d(\gamma(t), y) < (\delta - t) + t = \delta,$$

contradicting $d(p) = \delta$. Hence $r \geq \delta$.

For $t < r$, the segment $\gamma|_{[0,t]}$ minimises to X. Freeness of $B(\gamma(t), t)$ gives $d(\gamma(t)) \geq t$, while $d(\gamma(t)) \leq d(\gamma(t), x_p) \leq t$; hence equality throughout and $\gamma|_{[0,t]}$ realises $d(\gamma(t))$.

Case $r = \infty$. Then γ is a ray with $B(\gamma(t), t) \cap X = \emptyset$ for all $t > 0$. The identity

$$\bigcup_{t>0} B(\gamma(t), t) \;=\; \{b_\gamma < 0\}$$

holds by: ($\subset$) $b_\gamma(y) \leq d(y, \gamma(t)) - t < 0$; ($\supset$) if $b_\gamma(y) < 0$ then $d(y, \gamma(t)) < t$ for large t. Thus $\{b_\gamma < 0\}$ is an X-free open horoball containing p (since $b_\gamma(p) = -\delta < 0$), so $p \notin \widehat{X}$. This establishes the equivalent "non-reconstructible $\Rightarrow$ inside free horoball" form simultaneously.

Case $r < \infty$. Set $c := \gamma(r)$. By continuity, $d(c) = r$. Since $d(p, c) = r - \delta < r$, we have $p \in B(c, r)$ strictly. By construction $c \in \mathrm{Cut}(X)$, so Lemma 2.7 gives $c \in \overline{M_X}$. By continuity of $d(\cdot)$ and $d(p, \cdot)$, for any $\varepsilon < r - \delta$ there exists $a \in M_X$ near c with $d(p, a) < d(a)$, i.e. $p \in B(a, d(a)) \subset \mathcal{R}_X$. □

Corollary 3.2 (Compact manifolds)**.** *On a compact complete Riemannian manifold, X^c is reconstructible for every non-empty closed X.*

Proof. Compact M has no rays (geodesics are not globally minimising beyond the diameter), so $\widehat{X} = M$ and every $p \in X^c \subset \widehat{X} \setminus X$ is reconstructible by Theorem 3.1. □

Remark 3.3. Theorem 3.1 subsumes and simplifies Corollary 5 of [1]. It replaces Carathéodory's lemma and the simplex inflation of Proposition 2 with the cut-locus argument; the key estimate $r \geq \delta$ replaces the half-space contradiction.

4. Reconstructing defective horoballs on Hadamard manifolds

Throughout this section (M, g) is a *Hadamard manifold*: complete, simply connected, sec ≤ 0. Relevant properties used below.

(H1) b_ξ is $C^{1,1}$ with $|\nabla b_\xi| \equiv 1$ (Heintze–Im Hof).

(H2) From each $q \in M$ there is a unique ray σ_q asymptotic to a given ideal point $\xi \in \partial_\infty M$; σ_q is the flow line of $-\nabla b_\xi$ through q.

(H3) If σ_1, σ_2 are asymptotic to the same ideal point ξ, then $b_\xi \circ \sigma_1(\cdot)$ and $b_\xi \circ \sigma_2(\cdot)$ differ by an additive constant (follows from convexity of $t \mapsto d(\sigma_1(t), \sigma_2(t))$), Bridson–Haefliger II.8).

Theorem 4.1 (Defective horoball reconstruction)**.** *Let M be Hadamard, X closed, and $\mathcal{H}_\gamma$ a defective maximal X-free horoball with trapped witness w. Then $\mathcal{H}_\gamma \subset \mathcal{R}_X$.*

Proof. Normalise $b := b_\gamma - c_\gamma$ so that $X \subset \{b \geq 0\}$, $\mathcal{H}_\gamma = \{b < 0\}$, and $w \in \{b = 0\} \cap \widehat{X} \setminus X$. Let $\xi = \gamma(\infty)$. By (H2), let $\sigma : [0, \infty) \to M$ be the unique asymptotic ray from w, so $b(w_t) = -t$ for $w_t := \sigma(t)$. Since b is 1-Lipschitz and $b \geq 0$ on X:

$$(*) \qquad d(w_t) \ \geq \ -b(w_t) \ = \ t.$$

Step 1 (strictness). *$d(w_t) > t$ for all $t > 0$.*

Suppose $d(w_t) = t$ realised by a minimising segment τ from a foot $x_t \in X$ to w_t. Since $b(x_t) \geq 0$, $b(w_t) = -t$, and τ has length t, the 1-Lipschitz estimate forces $b(x_t) = 0$ and $b(\tau(s)) = -s$ for all $s \in [0, t]$.

Extend τ by the asymptotic ray σ_{w_t} at w_t. The concatenation P satisfies $b(P(s)) = -s$ for all $s \geq 0$ and $|\dot{P}| \equiv 1$, hence $\langle \nabla b, \dot{P} \rangle \equiv -1$. Since $|\nabla b| \equiv 1$ by (H1), this forces $\dot{P} = -\nabla b(P)$, so P is a unit-speed geodesic ray and equals the unique asymptotic ray from each of its points by (H2).

Both P and σ pass through w_t with velocity $-\nabla b(w_t)$, so they agree. Then $x_t = P(-t)_{\text{backward}} = w = \sigma(-t)_{\text{backward}}$ (both are at parameter $-t$ on the same geodesic), giving $x_t = w \notin X$ — contradiction.

Step 2 (upgrade to medial centres). *For each $t > 0$ there exists $c_t \in \overline{M_X}$ with $B(w_t, d(w_t)) \subset B(c_t, d(c_t))$.*

Let $x_t^* \in m(w_t)$ and γ_t the geodesic extension past w_t. Define r_t as in the proof of Theorem 3.1; by Step 1, $r_t \geq d(w_t) > t$. If $r_t = \infty$ then γ_t is a ray; since $b(w) = 0$ and by (H3) b and b_{γ_t} (normalised at w_t) differ by the constant $b(w_t) = -d(w_t)$:

$$b_{\gamma_t}(w) \ \leq \ b_{\gamma_t}(w_t) + d(w, w_t) \ = \ 0 + t \ = \ t,$$

but also $b_{\gamma_t}(w) \leq b(w) + d(w_t) - t = d(w_t) - t < 0$ by Step 1. So w lies in the open free horoball $\{b_{\gamma_t} < 0\}$, contradicting $w \in \widehat{X}$. Hence $r_t < \infty$, $c_t := \gamma_t(r_t) \in \mathrm{Cut}(X) \subset \overline{M_X}$, $d(c_t) = r_t$, and for any $y \in B(w_t, d(w_t))$:

$$d(y, c_t) \leq d(y, w_t) + (r_t - d(w_t)) < r_t = d(c_t).$$

Step 3 (swallowing). *Every $p \in \mathcal{H}_\gamma$ lies in $B(c_t, d(c_t))$ for large t.*

By property (H3), $b_\sigma = b + K$ for a constant K fixed by $b_\sigma(w) = 0 = b(w)$: so $b_\sigma = b$. The function $t \mapsto d(p, w_t) - t$ is non-increasing (triangle inequality along σ) and converges to $b_\sigma(p) = b(p) < 0$. Hence for all large t:

$$d(p, w_t) < t \leq d(w_t),$$

i.e. $p \in B(w_t, d(w_t)) \subset B(c_t, d(c_t))$. Approximate c_t from M_X to conclude $p \in \mathcal{R}_X$. $\square$

Theorem 4.2 (Sufficiency on Hadamard manifolds)**.** *Let M be Hadamard and X closed. Then*

$$\big(\widehat{X} \setminus X\big) \ \cup \ \bigcup \big\{ \mathcal{H}_\gamma : \mathcal{H}_\gamma \text{ maximal free and defective} \big\} \ \subseteq \ \mathcal{R}_X.$$

In particular X^c is reconstructible if and only if $X^c \subseteq \widehat{X} \cup \bigcup_{\text{defective}} \mathcal{H}_\gamma$.

Proof. The first inclusion follows from Theorems 3.1 and 4.1. The second statement uses that a set is reconstructible iff every point is reconstructible. $\square$

5. Necessity: partial result and open gap

Conjecture 5.1. *On a Hadamard manifold, equality holds in Theorem 4.2: a point $p \in X^c$ is reconstructible if and only if $p \in \widehat{X} \setminus X$ or p belongs to some defective maximal free horoball.*

The following partial result shows that reconstructible points outside $\widehat{X}$ are *witnessed by* defective horoballs, but does not yet place those points inside one.

Proposition 5.2 (Partial necessity)**.** *Hadamard M, $p \in \mathcal{R}_X \setminus \widehat{X}$ witnessed by $a \in M_X$, $q \in m(a)$. Let w be the first point of the geodesic segment $[p, q]$ in $\widehat{X}$ (it exists since $\widehat{X}$ is closed and $q \in X \subset \widehat{X}$). If $w \neq q$, then $w \in \widehat{X} \setminus X$ and some maximal free horoball is defective with trapped witness w.*

Proof sketch. Geodesic convexity of $B(a, d(a))$ (on Hadamard manifolds, balls are convex) gives $[p, q) \subset B(a, d(a)) \subset X^c$, so $w \notin X$. Points $w_k \to w$ on $[p, w)$ lie in X^c, so none are in $\widehat{X}$; each w_k is thus contained in some X-free open horoball $\mathcal{H}_\gamma[\xi_k]$. Pass to a convergent subsequence $\xi_k \to \xi_*$ in $\partial_\infty M \cong S^{n-1}$ (compact); the Busemann functions converge locally uniformly. The function $\xi \mapsto c_\xi$ (appropriately normalised) is upper semicontinuous (evaluating at any $x \in X$ gives $c_{\xi_k} \le b_{\xi_k}(x) \to b_{\xi_*}(x)$, so $\limsup c_{\xi_k} \le c_{\xi_*}$). Hence $b_{\xi_*}(w) \le \limsup c_{\xi_k} \le c_{\xi_*}$; the reverse inequality holds since $w \in \widehat{X}$, giving $b_{\xi_*}(w) = c_{\xi_*}$: w lies on the horosphere of a maximal free horoball, and $w \notin X$, so that horoball is defective. □

The gap. The argument shows that some defective horoball $\mathcal{H}_\gamma[\xi_*]$ has w on its boundary, but it does not show $p \in \mathcal{H}_\gamma[\xi_*]$. Convexity of b_{ξ_k} along the segment $[p, w]$ bounds the function from *below* at the endpoint p, not from above. Two sub-cases also require additional argument:

- $w = q$ (the segment meets $\widehat{X}$ only at the foot);
- the entry is tangential (the segment is tangent to $\partial\widehat{X}$ at w), in which case the limiting direction ξ_* may not separate p from $\widehat{X}$.

We note that *the same gap exists in the original Euclidean proof*; see §7.

6. Curvature and hypothesis audit

- **Theorem 3.1.** Only completeness is used. No curvature, no simple connectedness.
- **Theorem 4.1.** The Hadamard assumption is used twice: (H1) (smooth Busemann with $|\nabla b| \equiv 1$) in Step 1, and (H3) (asymptotic rays differ by constants) in Step 3. (H2) (uniqueness of co-rays) is also used in Step 1.
- **Proposition 5.2.** Ball convexity and visual boundary compactness are used; both hold for Hadamard manifolds.
- Outside nonpositive curvature, Step 1 is the fragile point: with conjugate points, two distinct rays through w_t can both be asymptotic to ξ while descending b at unit rate, breaking the uniqueness argument.

Examples outside Hadamard. Let $X = S^1 \times \{0\} \subset S^1 \times \mathbb{R}$ (flat cylinder). Here $M_X = \emptyset$ (unique foot for every point not on X), $\mathcal{R}_X = \emptyset$; rays are vertical, the two horoballs are the open half-cylinders, $\widehat{X} = X$, no defects — the theorem predicts $\mathcal{R}_X = \emptyset$, which is correct.

On a paraboloid of revolution with $X = \{\text{vertex}\}$: the vertex is a pole, $M_X = \emptyset$, $\mathcal{R}_X = \emptyset$; every $q \neq$ vertex lies in the free horoball of its own meridian ray ($b_\gamma(q) < 0$ while $b_\gamma(\text{vertex}) = 0$), so $\widehat{X} = X$, no defects, prediction $\mathcal{R}_X = \emptyset$ is correct. Both examples are consistent with necessity too.

7. Audit of the original Euclidean paper

Working at the horoball level exposes three issues in [1].

1. Proposition 3, final step. The last paragraph claims: "$\|c_n\| \to \infty$, $c_n/\|c_n\| \to \eta$ implies $d(c_n) > \|p - c_n\|$ for n large." This is false as a general implication. Consider centres at height $H_n \sim n$ with transverse drift $T_n \sim n^{0.9}$; the direction converges to η yet $T_n^2 \gg H_n$, and the ball around c_n does not swallow a fixed point p at height h. The proof *does* survive because the ε-pinning of feet via upper semicontinuity of m keeps $\|p - x_n\|$ bounded throughout. The published last step skips the tangency inequality

$$2r_n\langle v_n, p - x_n\rangle > \|p - x_n\|^2$$

that actually closes the argument. The result is correct but the proof is incomplete as written.

2. Proposition 4, quantitative gap. The feet in Proposition 4 drift transversally up to $\sqrt{2\delta t}$, so $\|p - x_t\|^2 \sim 2\delta t$. With only $r_t \ge t$ available, the tangency inequality closes for heights $h > \delta$ but not for $0 < h \le \delta(w)$. A fix is available when $X \cap L \neq \emptyset$: move the trapped witness w along the contact face toward $X \cap L$ to drive $\delta(w) \to 0$, then take a union. When $X \cap L = \emptyset$ that route fails.

Notably, the proof of Theorem 4.1 above, restricted to $\mathbb{R}^n$, gives a shorter and complete proof of Propositions 3–4 combined: Step 1 (three lines using b-calculus) replaces the directional-radius induction, and Step 3 gives the swallowing without tracking feet at all.
3. conv X vs. $\overline{\text{conv}}X$. The defect condition in Theorem 6 uses the non-closed hull conv X, while Corollary 5 and the entry-point in the necessity proof live in the closure $\overline{\text{conv}}X$. The statement and all proofs should be stated uniformly with $\overline{\text{conv}}X$; the horoball framework produces the closure version automatically (as the intersection of closed superlevel sets).
The necessity gap (common to both settings). The necessity proof in [1] (last paragraph of Theorem 6) picks a hyperplane L that "separates p and conv X" and assumes it passes through the entry point w with p *strictly* in L^+. At a tangential entry (the segment $[p, q]$ is tangent to $\partial\overline{\text{conv}}X$ at w), no such L exists passing through w, and a translating separating hyperplane may touch the hull inside X, where it is useless. The same case is open in our Hadamard version (end of §5). Plausible repairs in $\mathbb{R}^n$: perturb q within $m(a)$, or induct on dimension inside the offending face.

8. Summary and open problems

Result	Hypothesis	Status
Thm 3.1: $\widehat{X} \setminus X \subset \mathcal{R}_X$	Complete	Proved
Cor 3.2: X^c reconstructible	Compact	Proved
Thm 4.1: defective horoballs $\subset \mathcal{R}_X$	Hadamard	Proved
Prop 5.2: witnesses exist	Hadamard	Proved
Conj 5.1: converse of Thm 4.2	Hadamard	Open
Necessity in [1]	$\mathbb{R}^n$	Gap (see §7)

Open problems:

(1) Close the necessity gap (both in $\mathbb{R}^n$ and on Hadamard manifolds). A candidate approach: if $p \in B(a, d(a))$ with a having two feet q_1, q_2, show directly that the horoball in the direction from the midpoint of $\overline{q_1 q_2}$ toward a is defective and contains p.
(2) Extend Theorem 4.1 beyond nonpositive curvature by replacing (H1)–(H3) with weaker visibility or no-conjugate-point conditions.
(3) Formulate and prove the λ-reconstructibility analogue: replace horoballs with λ-truncated horoballs (balls of radius $\geq \lambda$ whose centres lie in $M_\lambda(X)$).
(4) Determine whether $\mathcal{R}_X$ always equals the horoball hull minus the defect-free boundary, without any curvature assumption.

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